# TxBench-PP: Verifiable Evaluation of AI Agents on Preclinical Pharmacology

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**ABSTRACT**

We introduce TxBench-PP, a verifiable benchmark for evaluating AI agents on small-molecule preclinical pharmacology. TxBench-PP is the first focused slice of a broader TherapeuticsBench effort organized across drug-discovery stages and therapeutic modalities. It targets recurring cluster-style decisions: whether agents can recover decision-relevant conclusions from supplied assay evidence rather than recalling drug labels, ranking by a single metric, or applying generic checklists. The current task inventory maps 70 evaluations to a nine-stage Assays Roadmap. Seven stages have coverage, with strongest current coverage in causal target validation and mechanism-of-action / pharmacodynamic reasoning, single-source coverage in screening, pharmacogenomics, safety, and translational efficacy, and a near-empty compound-target characterization stage. Agents receive realistic workflow snapshots, inspect files in a coding environment, and return structured answers graded deterministically. [**TODO:** Replace this final paragraph with model-run results: number of evaluations, model-harness pairs, trajectories, top pass rates, and the main failure pattern.]

## TxBench-PP in TherapeuticsBench

TxBench-PP is the current small-molecule preclinical pharmacology slice of a broader TherapeuticsBench roadmap. The full benchmark family is organized by drug-discovery stage and therapeutic modality; this paper focuses on repeated, verifiable cluster decisions in small-molecule programs. These decisions sit between initial discovery and clinical development: whether a molecule engages the intended biology, whether the evidence supports its mechanism, whether exposure can cover the relevant potency range, and whether safety signals block advancement.

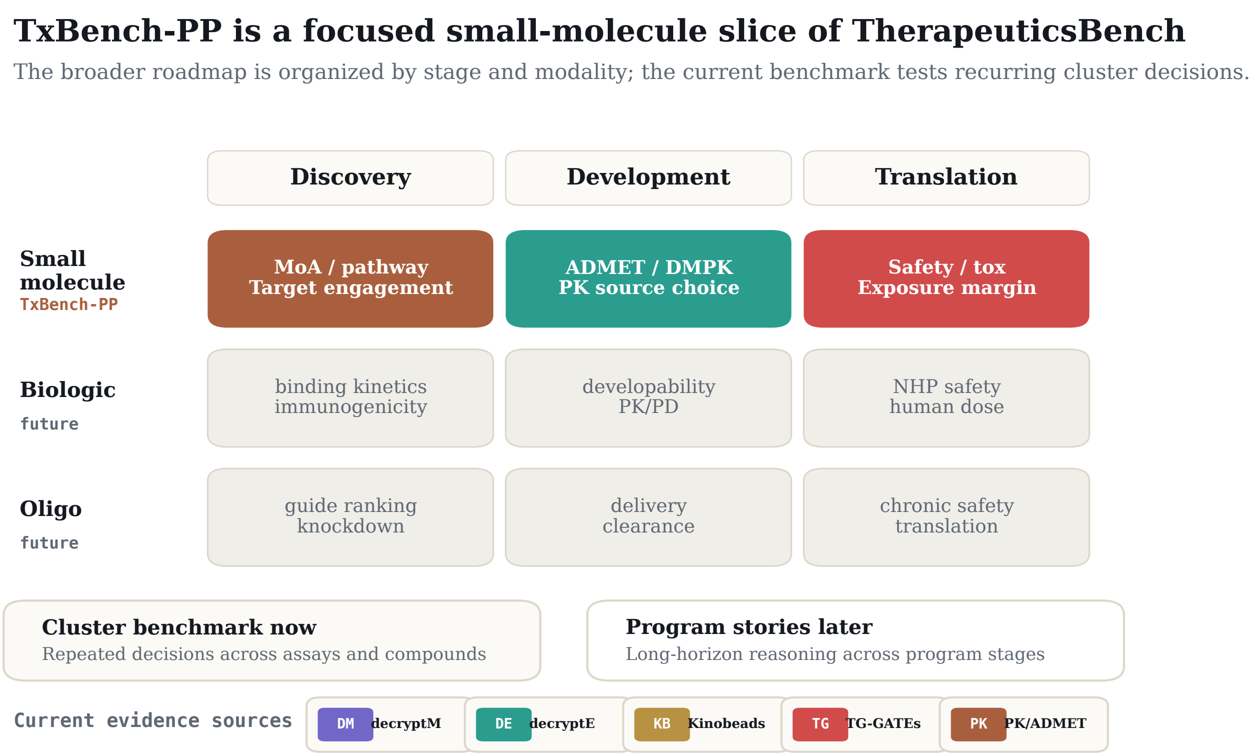



**Figure 1: TxBench-PP as a TherapeuticsBench slice.** TherapeuticsBench is intended to span discovery, development, and translation across small molecules, biologics, and oligonucleotides. TxBench-PP covers the small-molecule preclinical pharmacology slice as a cluster benchmark: repeated local decisions across assay families and compounds. Program-story benchmarks that follow one therapeutic program across discovery, development, and translation are reserved for future work.

# Introduction

In preclinical drug discovery, experiments are slow and expensive, but many program-limiting decisions are interpretive: which signal is real, which compound should advance, and which liability should block a program. Preclinical pharmacology converts assay evidence into these decisions. A compound can have an attractive potency value and still be a poor candidate because the apparent mechanism is an artifact, the engaged protein is a complex contaminant, the safety signal is protocol-specific, or total exposure does not translate into unbound target coverage.

AI agents can inspect files, write analysis scripts, and invoke scientific software, but their conclusions often remain vulnerable to plausible shortcuts. In drug discovery, those shortcuts have familiar forms: importing the known mechanism of a blinded compound, trusting the largest apparent effect without quality control, ranking by potency or total exposure alone, or applying a textbook safety threshold outside the protocol where it was defined.

We introduce TxBench-PP, a benchmark for small-molecule preclinical pharmacology decisions. It should be read as a focused cluster benchmark within the broader TherapeuticsBench roadmap rather than as a complete test of drug discovery. Each evaluation supplies the agent with realistic analysis artifacts from one stage of the preclinical pharmacology evidence chain and asks for a structured, deterministically gradable conclusion. The benchmark tests whether an agent can recover the result supported by the provided evidence, not whether it can recall a drug class, run a default workflow, or optimize for a single number.

The current benchmark maps 70 evaluations to a nine-stage Assays Roadmap. Coverage is strongest in causal target validation and mechanism-of-action / pharmacodynamic reasoning; the clearest build gap is compound-target characterization. The underlying sources still matter: several covered stages are dominated by a small number of datasets, so diversifying assay sources is a central part of the benchmark roadmap. **[TODO:** Add final run summary once result aggregation is complete.]

# Benchmark Construction

To construct TxBench-PP, we decompose preclinical pharmacology workflows into short, gradeable scientific decisions. Examples include identifying a pathway-coherent mechanism from dose-resolved phosphoproteomics, separating true chemoproteomic target engagement from complex contaminants, reconciling conflicting clearance measurements, and integrating blood chemistry, pathology, and survival into an in vivo safety call.

This makes TxBench-PP a cluster-style benchmark: the same kind of decision is tested repeatedly across assay contexts, compounds, and evidence sources. Future TherapeuticsBench story evaluations can test the complementary skill of following one therapeutic program across discovery, development, and translation. TxBench-PP instead asks whether agents can make the local scientific decisions that those longer program stories depend on.

Each task includes a snapshot of assay outputs immediately before a target decision, metadata needed to interpret the files, a high-level task description, and a deterministic grader. Tasks specify what result should be recovered rather than prescribing the exact analysis method. Procedural details are included only when they are part of the scientific decision, such as whether free or total exposure should be compared to a pharmacologic threshold.

The benchmark follows three design criteria. First, tasks must be verifiable: answers are checked by structured label matching, numerical tolerances, rank-order checks, or all-of field comparisons. Second, tasks should be decision-relevant: the answer should correspond to a real go/no-go or prioritization judgment in preclinical pharmacology. Third, tasks should resist shortcuts: agents should need to inspect the supplied evidence, account for quality issues, and avoid relying on memorized drug labels or generic checklists.

We separately track decision-blocking gaps: issues that should change the program decision if recognized, such as a tox-species mismatch, insufficient free-drug coverage, a hook effect, a contaminated chemoproteomic hit, or survivor bias in pathology sampling. These labels are intended to support diagnostic analysis after endpoint grading.

# Evaluation Inventory

The current suite is organized around the Assays Roadmap rather than around source datasets. Each evaluation is assigned to the roadmap stage that best matches its primary biological skill. The current inventory includes 70 evaluations across seven of the nine stages. S1 disease/model characterization and S4 human genetic target support are deferred for this release; S4 is better handled by GenomicsBench.

The roadmap view changes the benchmark story. Coverage is broad by stage count, but narrow by dataset diversity. Causal target validation and MoA/PD are ready for v1. Screening, pharmacogenomics, safety, and translational efficacy have useful coverage but are single-source dominated. The largest in-scope gap is S7 compound-target characterization: binding, potency, selectivity, and cellular target engagement are almost absent.

## Evaluation inventory follows the nine-stage Assays Roadmap

Coverage is broad by stage count, but several covered stages remain single-source dominated.

**70** evals | **7/9** stages covered | **S7** build gap

| Stage | Evals | Status | Roadmap interpretation |
|---|---|---|---|
| S1 Disease / model characterization | 0 | Deferred | out of v1 scope |
| S2 Primary screening & hit confirmation | 7 | Diversify | PRISM-heavy |
| S3 Drug response / pharmacogenomics | 11 | Diversify | mostly PRISM |
| S4 Human genetic target support | 0 | Deferred | GenomicsBench |
| S5 Causal target validation | 12 | Ready | v1 coverage |
| S6 MoA / pharmacodynamics | 17 | Ready | strongest coverage |
| S7 Compound-target characterization | 1 | Build | largest gap |
| S8 Developability & safety | 16 | Diversify | cardiotox-heavy |
| S9 Translational efficacy | 6 | Diversify | one NSCLC source |

Counts and statuses are from the Assays Roadmap coverage mapping. S1 and S4 are intentionally deferred for this release.

**Figure 2: Evaluation inventory by Assays Roadmap stage.** Current TxBench-PP evaluations are mapped to the roadmap stage that best captures the biological skill being tested. The inventory is broad across stages but uneven across assay families: S5 and S6 are ready for v1, S7 is the clearest build gap, and S2, S3, S8, and S9 need source diversification.

**Table 1: Roadmap-stage inventory.** Counts are from the Assays Roadmap coverage mapping of 70 current evaluations. "Ready" stages are sufficient for v1; "Diversify" stages have useful but source-narrow coverage; "Build" marks the largest missing pillar.

| Roadmap stage | Evals | Status | Assay emphasis and next gap |
|---|---|---|---|
| S1. Disease / model characterization | 0 | Deferred | Model subtype, expression, mutation/CNV context, pathway state, IHC/IF, RNAscope, and model QC are out of the current build queue. |
| S2. Primary screening / hit confirmation | 7 | Diversify | PRISM-heavy. Add biochemical or reporter HTS, Z-prime and signal-window plate QC, counterscreens, and frequent-hitter filtering. |
| S3. Drug response / pharmacogenomics | 11 | Diversify | Mostly PRISM. Add GDSC, CTRP, CCLE, Beat AML, organoid screens, single-cell perturbation, and censored-curve handling. |
| S4. Human genetic target support | 0 | Deferred | GWAS, QTL, colocalization, Mendelian randomization, and tissue-of-action matching are assigned to GenomicsBench. |
| S5. Causal target validation | 12 | Ready | Strong v1 coverage. Optional expansion: Perturb-CITE-seq, rescue assays, variant-function classification, and FACS sort-bin screens. |
| S6. MoA / pharmacodynamics | 17 | Ready | Strong v1 coverage. Optional expansion: LINCS/CMap, DRUG-seq, Seahorse, cytokine panels, reporter dose-response, and washout PD. |
| S7. Compound-target characterization | 1 | Build | Highest-priority gap. Add ChEMBL/BindingDB potency curation, selectivity panels, CETSA/NanoBRET, Kinobeads, and biochemical-to-cellular potency comparison. |
| S8. Developability / safety | 16 | Diversify | Cardiotox dominated. Add measured hERG, DILI, Ames/micronucleus, permeability, metabolic stability, CYP/DDI, Tox21/ToxCast, and therapeutic-index reasoning. |
| S9. Translational efficacy | 6 | Diversify | Single NSCLC source. Add organoid/patient concordance, PDX pharmacology, immune co-culture, ctDNA, responder-fraction, and exposure-response PK/PD. |

**Cross-stage gap.** The roadmap also identifies a missing class of end-to-end go/no-go evaluations: tasks that require an agent to weigh efficacy, selectivity, target engagement, exposure, and safety together. The current inventory covers many of these component decisions, but future TxBench-PP tasks should test multi-stage synthesis across at least three evidence blocks.

# Results

## No agent reliably recovers preclinical pharmacology decisions

We evaluated 16 model–harness configurations, comprising 11 models across three agent harnesses, on 100 preclinical pharmacology tasks. Each configuration was run three independent times per task, yielding 4,800 agent trajectories in total. A trajectory was scored as passing only if its submitted answer satisfied every task-specific grader. We report pass rate as the fraction of passing trajectories, with 95% confidence intervals computed across the 100 tasks.

The four strongest configurations were Claude Opus 4.8 on Pi, with a pass rate of 59.3% (178/300; 95% CI, 51.0–67.6); GPT-5.5 on Pi, 55.3% (166/300; 47.0–63.6); Claude Opus 4.8 on Claude Code, 54.7% (164/300; 45.9–63.4); and Gemini-3.5-Flash on Pi, 51.3% (154/300; 42.9–59.8). These systems did not separate into a clear winner. Instead, the top configurations formed a broad, overlapping performance tier, with no configuration rising decisively above the 50–60% range. Even the best system left more than 40% of decisions unrecovered.

Thus, across providers, model generations, and agent harnesses, no system approached the reliability required for preclinical decision-making. The absence of a clear leader, together with a performance ceiling near 60%, suggests that current agent systems do not reliably recover preclinical pharmacology decisions.

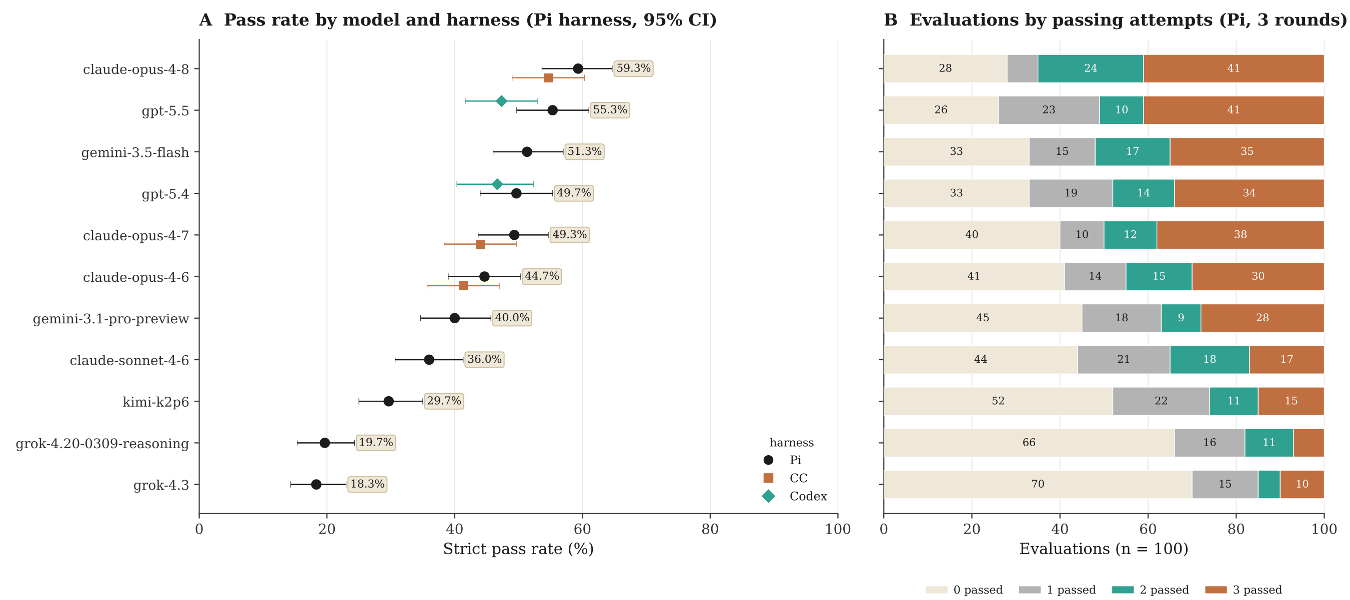



**Figure 3: Pass rates and reliability across model–harness configurations.** (**A**) Strict pass rate for each of the 16 model–harness configurations across all 4,800 agent trajectories, with 95% bootstrap confidence intervals computed over the 100 tasks. Each trajectory was scored pass only when its submitted answer satisfied every task-specific grader. Alternate harnesses (Claude Code, Codex) are shown as offset points for the models that ran them. (**B**) Task-level reliability for the Pi harness, partitioning the 100 evaluations for each model by how many of the three independent rounds passed.

## Most failures map to a compact failure-mode taxonomy

To understand why agents fail, we manually reviewed every failing (model, evaluation) run on the Pi harness and labeled each against the evaluation's documented scientific intent, yielding 1,834 labeled failures. We classify each failure along two complementary axes: the *scientific error*, which captures what is wrong with the answer, and the *behavioral cause*, which captures how the model arrived there (Figure 4).

The scientific axis comprises five recurring error families (Figure 4A, restricted to the $n = 1{,}436$ failures that contain a scientific error). The largest, flawed method, resolves into two main sub-modes and a small tail. A *skipped-QC* error (17%) omits a required preprocessing or quality-control step: in one case, a model aggregated CRISPR guides to gene-level scores without first dropping guides that map to multiple loci, so paralog-family artifacts flooded the essential-gene list. A *wrong-reference-frame* error (21%) runs the pipeline correctly but compares against the wrong baseline, comparator, or normalization. The small *wrong-assay-layer* tail (2%) reads the wrong readout from the right experiment. On a proteasome-kinetics task asking which compound degrades substrate faster, the answer lives in the ubiquitin-remnant layer of a dual-readout mass-spectrometry run. A model that answers from the phosphoproteome layer of the same run reaches a different conclusion on what is a different question.

Beyond flawed method, four families remain. A *calibration* error (31%) draws a cutoff a domain expert would set differently. For instance, in one competition-binding panel, models carried forward all 33 nonzero kinases when only the 8 strong binders above a clear affinity gap were supported. A *perception* error (10%) misreads a raw image, such as counting apoptotic debris as viable organoids or mis-segmenting two-channel colocalization. *Prior over data* (10%) answers from textbook or famous biology against the staged evidence: one model computed the strongest data-supported biomarker—a vanadium compound correlated with SLC26A2 at $r = -0.48$—then discarded it for the textbook elesclomol-FDX1 cuproptosis pair. *Over-reading* (10%) states a stronger claim than the data supports, such as reporting tumor regression from a sub-baseline median when the per-animal measurements show only growth arrest. Together, method and calibration account for 71% of scientific errors.

The behavioral axis records how each failure happened, across all $n = 1{,}834$ runs (Figure 4B). Two-thirds (67%) are *engaged, analytically wrong*: the model did the work in good faith and made an analytical error. The remaining third splits into one evidence-handling failure and three process failures. *Prior-driven override* (6%), the lone evidence-handling case, engages the data and often computes the supporting result, then defers to a textbook or famous-biology prior. *Weak commitment* (9%) performs the right analysis but submits a final answer that does not match it, a mismatch that appears even in the strongest models and points to a missing final-answer check. *Abandoned-correct* (12%) has a passing result within reach and then commits to a worse, more canonical one such as a published figure or standard global method. Like prior-driven override, it occurs most often in the longest runs. *Under-investment* (6%) reaches an answer with little tool use, submitting a plausible call before running the per-dose computation or QC the task requires.

The two axes distribute differently across models (Figure 4C,D). The behavioral mix varies by model. Under-investment is concentrated in the two weakest models, at 32% of grok-4.3's failures and 17% of grok-4.20-reasoning's against 0–1% for the others. Abandoned-correct is spread across the frontier at 7–17%, highest in gpt-5.5 and gemini-3.1-pro (17% each). The scientific-error mix is nearly constant, with method and calibration accounting for 58–77% of every model's scientific errors regardless of pass rate.

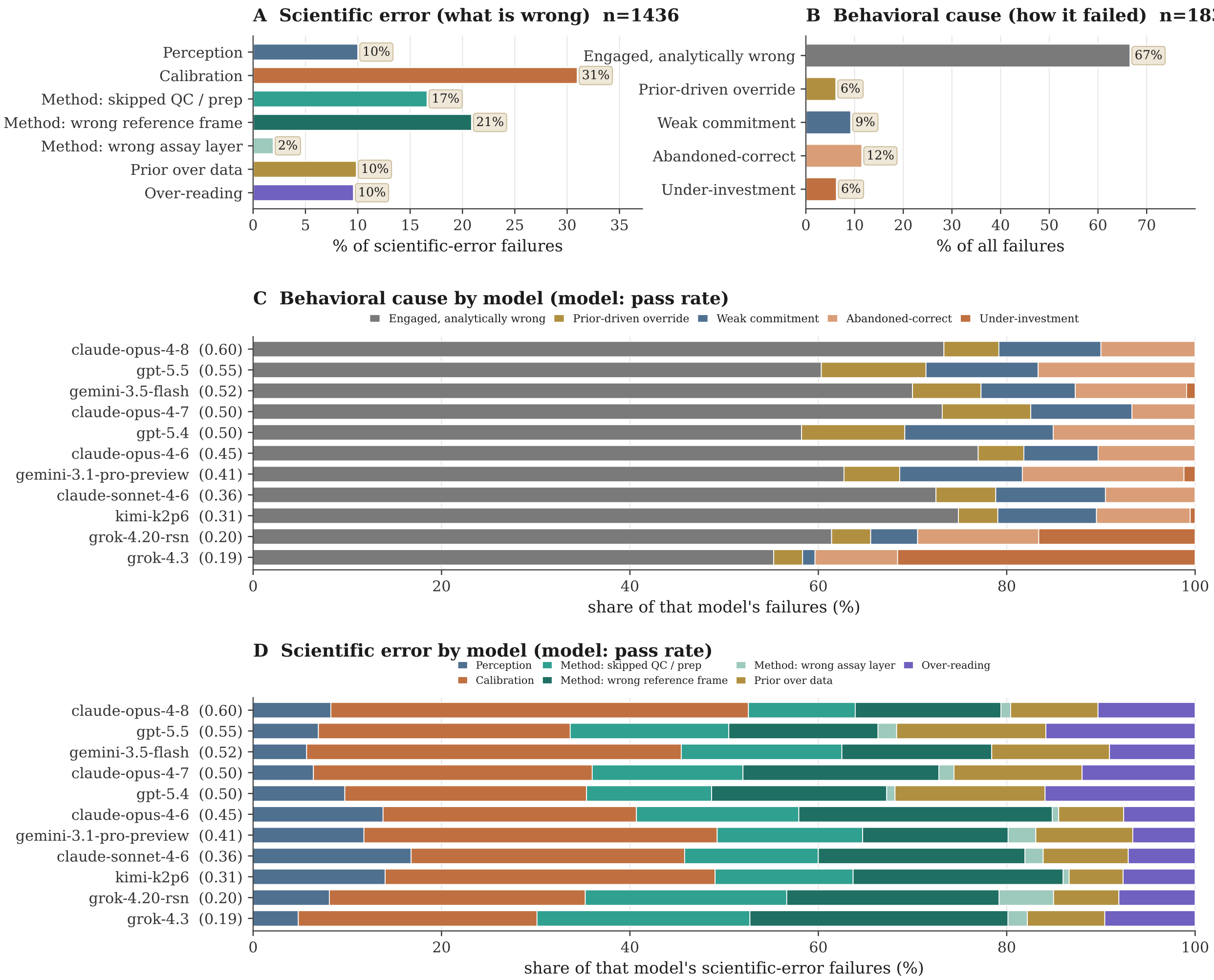



**Figure 4:** **A two-axis failure taxonomy.** All 1,834 failing (model, evaluation) runs on the Pi harness, labeled against each evaluation's scientific intent and classified along two complementary axes. (**A**) The scientific-error axis (what is wrong with the answer) is dominated by method and calibration errors (71% combined), which are general analytical skills rather than domain knowledge; the method family resolves into a skipped-QC/preprocessing sub-mode and a larger wrong-reference-frame sub-mode. (**B**) The behavioral-cause axis (how the model failed) shows that a third of failures are non-analytical: a prior-driven override of the model's own evidence, plus three process breakdowns (weak commitment, abandoned-correct, under-investment). (**C**) The behavioral mix is a model signature: under-investment dominates only the weakest model, while the strongest models fail almost entirely through engaged, analytically wrong reasoning. (**D**) The scientific-error mix, by contrast, is largely model-invariant: method and calibration errors dominate every model regardless of overall accuracy, so the *kinds* of scientific mistakes are shared across the frontier even as their rate falls (each bar normalized within that model's scientific-error failures). A scientific-error × behavioral-cause cross-tabulation is provided in the supplement (`fig_fm_crosstab`).

## The dominant failure mode shifts characteristically across pipeline stages

Pass rates range from 27% at the hardest stage to 55% at the easiest stage (Figure 5). Walking the arc in order shows that each stage has characteristic strengths and weaknesses. We look at a representative easy and hard evaluation for each stage, with the fraction of passing runs in parentheses.

Overall, the ordering of models at each stage closely mirrors their overall performance (Figure 5C; mean Spearman $\rho = 0.80$ across stages, ranging from 0.56 at S7 to 0.92 at S8), with the two Grok models ranking lowest at every stage. Three features stand out against this baseline. The two highest cells in the matrix are gemini-3.5-flash and gpt-5.5 at drug response, at 87% (95% CI 70–95, $n = 30$) and 86% (69–95, $n = 29$), well above their own overall rates of 52% and 55%. The clearest stage effect is within a single model. gemini-3.5-flash falls from 87% at S3 to 28% (16–44, $n = 36$) at causal target validation. The widest within-family spread is at hit screening, where claude-opus-4-8 scores 50% (31–69, $n = 24$) and the older claude-opus-4-6 12% (4–31, $n = 24$).

*Primary screening and hit prioritization* (27%) is the hardest stage. Models are competent at the mechanical steps, such as robustly collapsing noisy replicates (`PRISM05`, 58%). They fail by choosing the wrong reference frame for QC. On outlier-pool QC (`PRISM01`, 18%) they flag whole well-by-pool groups by an aggregate median residual and exclude most of the cytotoxic positive-control wells, discarding real biology as a technical artifact. On an oncology drugs repurposing prioritization task (`PRISM_S2_11`, 0%) they rank by breadth of killing and nominate a pan-cytotoxic compound rather than a selective hit.

*Drug response and pharmacogenomics* (55%) is one of the easier stages but exposes a consistent weakness in unsupervised mechanism recovery. Models do well on direct evidence checks, such as separating a causal mediator from a correlated biomarker (`PRISM_S4_06`, 76%). They fail when asked to cluster compounds by mechanism from their response profiles. Across the embedding tasks they use a potency-dominated representation, a linear PCA or a Euclidean metric on raw profiles, so the leading component encodes potency and the intended drug class never separates from the rest. The same error recurs across statin collapse (`PRISM_UMAP_01`, 12%), oncology MoA recovery (`PRISM_UMAP_04`, 48%), and glucocorticoid substructure (`PRISM_UMAP_03`, 69%), so it is a property of the stage rather than a single task.

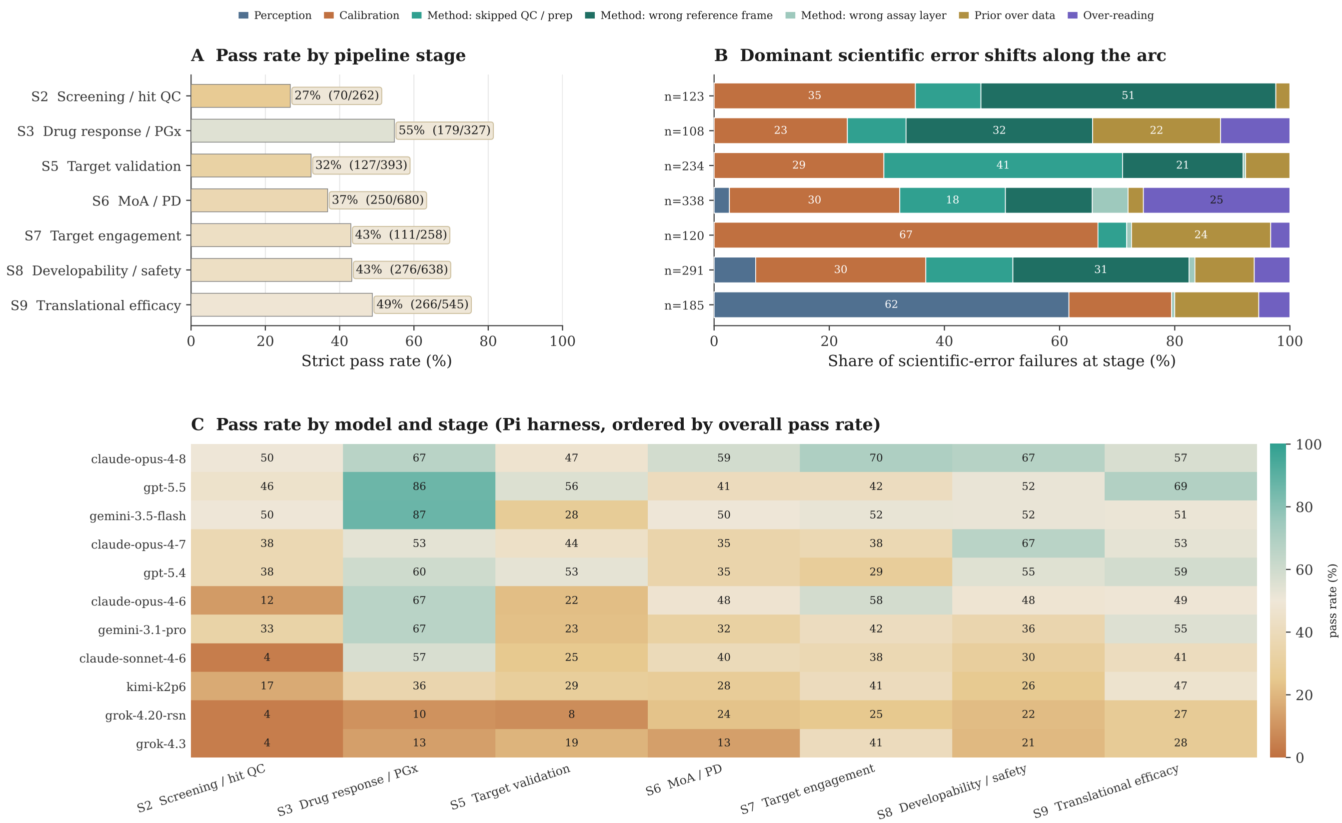
*Causal target validation* (32%) is the second-hardest stage and the one that most rewards data hygiene and statistical rigor. Models recover genuine dependencies from a clean screen (`HIT_05` replicate discordance, 69%). They routinely skip the library hygiene that CRISPR screens require. On multi-target guide contamination (`GG_01`, 16%) they aggregate guides to gene level without dropping multi-locus sgRNAs, so paralog-family false positives flood the dependency list. On the selective-set call (`SEL_01`, 6%) they ship thousands of genes behind a count-only filter with no effect-size floor.

*Mechanism of action and PD* (37%) is the largest stage, with 21 evaluations. Models reliably recover known program structure, such as assigning cells to the correct cell-cycle phase (`clust_04`, 84%). Their recurring failure is over-reaching on the mechanism, treating marginal or partial evidence as a strong, broad effect. On the antiinfective task (`ANTIINF_01`, 0%) they read near-noise pathway scores ($p \approx 0.05$) as genuinely engaged mechanisms. On the mTOR fingerprint (`TXBX_DM_v15E`, 6%) they correctly call mTOR the primary target but then count phosphosites from too broad a substrate panel, sweeping in adjacent AKT-axis sites instead of restricting to the canonical mTOR set, which inflates the affected-site count to almost twice what the data supports (59 versus 31).

*Compound-target engagement* (43%) asks whether a compound truly hits its target and how selectively, and the bottleneck is calibration, not detection (67% of the stage's scientific errors). Models detect the binding signal reliably, for example confirming a clean direct target by thermal shift (`thermal_shift_shared_direct_target`, 91%). What they lack is a principled cutoff for how much engagement counts, so they err in both directions. On a kinase-binding panel (`TXBX_KB_v11_DA1`, 0%) they accept every weak binder as a real hit, and on a control task (`CTRL01`, 21%) they invent a stringency cutoff the data never justified and discard five genuinely engaged targets.

*Developability and safety* (43%) depends on judging a signal against the right reference, and that is where models slip. They make a confident call when the standard is unambiguous, such as the hepatotoxicity verdict (`TXBX_TG_v11_A`, 91%). On the hERG safety-margin task (`de_14`) they size the margin against total drug concentration rather than the free, unbound fraction that actually reaches the target, overstating the safety window. The same habit appears on a cardiotoxicity task (`de_12`, 0%). The pathways most associated with a three-level toxicity-concern score must be found with a rank test across all three levels, but models collapse the score to a yes/no label and rank pathways by a binary test, so their top pathways diverge from the correct set and they miss the potassium-channel mechanism the data supports.

*Translational efficacy* (49%) leans on reading microscopy images, and that reading, not the downstream judgment, is the bottleneck (perception is 62% of its scientific errors). Models do well when the evidence is a table, such as calling a genotype-by-treatment rescue (`HAI2027_MM_FIG6C`, 89%). When they must score an image they misjudge what they see, and because the image is the last readout before a go/no-go, the misread becomes the wrong decision. On a two-marker immunostain (`HAI2027_MM_FIG2E`, 42%) they read faint, near-vehicle single-agent staining as a real signal and credit each monotherapy with activity, endorsing a single-agent program when only the drug combination is active. The organoid viability readouts are the hardest tasks in the benchmark (as low as 6%). When apoptotic debris is counted as living organoids (`organoid_moa_cytotoxic`), the mechanism call inverts, and a compound that is in fact killing the tumor cells is reported as inert or growth-promoting and advanced on that false basis.




| | S2 Screening / hit QC | S3 Drug response / PGx | S5 Target validation | S6 MoA / PD | S7 Target engagement | S8 Developability / safety | S9 Translational efficacy |
|---|---|---|---|---|---|---|---|
| claude-opus-4-8 | 50 | 67 | 47 | 59 | 70 | 67 | 57 |
| gpt-5.5 | 46 | 86 | 56 | 41 | 42 | 52 | 69 |
| gemini-3.5-flash | 50 | 87 | 28 | 50 | 52 | 52 | 51 |
| claude-opus-4-7 | 38 | 53 | 44 | 35 | 38 | 67 | 53 |
| gpt-5.4 | 38 | 60 | 53 | 35 | 29 | 55 | 59 |
| claude-opus-4-6 | 12 | 67 | 22 | 48 | 58 | 48 | 49 |
| gemini-3.1-pro | 33 | 67 | 23 | 32 | 42 | 36 | 55 |
| claude-sonnet-4-6 | 4 | 57 | 25 | 40 | 38 | 30 | 41 |
| kimi-k2p6 | 17 | 36 | 29 | 28 | 41 | 26 | 47 |
| grok-4.20-rsn | 4 | 10 | 8 | 24 | 25 | 22 | 27 |
| grok-4.3 | 4 | 13 | 19 | 13 | 41 | 21 | 28 |

**Figure 5: The dominant failure mode shifts across pipeline stages.** Failing Pi-harness runs placed along the preclinical pharmacology arc. S1 disease-model and S4 target-genetics stages are omitted, since S1 is backed by too few evaluations and S4 is deferred to GenomicsBench. (**A**) Strict pass rate by stage. The two lowest are the early data-handling stages S2 (screening/QC) and S5 (causal target validation). (**B**) Scientific-error composition by stage, restricted to failures carrying a scientific error and using the same seven categories as Figure 4A. The dominant error shifts characteristically along the arc, from wrong reference frame at screening (S2) to skipped QC at target validation (S5), calibration at target engagement (S7), and perception at the image-heavy translational stage (S9), while calibration persists as a thread at every stage. (**C**) Strict pass rate for every model at each stage, ordered by overall pass rate. The overall ordering is largely preserved across stages, with two robust rearrangements: gemini-3.5-flash and gpt-5.5 lead at drug response (S3), and gemini-3.5-flash drops sharply at causal target validation (S5).

## Some task categories remain far harder than others

Difficulty depends more on the type of judgment a task requires than on its biological domain. Grouping the 100 evaluations by task type and restricting to the 14 categories with at least four evaluations (86 of the 100 evaluations), the mean Pi-harness pass rate across the 11 models spans a 4.1-fold range, from 71% on exposure and PK tasks to 19% on hit prioritization and 17% on cheminformatics (Figure 6). Performance separates by the structure of the required answer rather than by subject matter. The four highest categories, exposure-PK (71%), biomarker discovery (56%), safety assessment (55%), and target engagement (52%), each resolve to a determinate target and average 58%. In each, the answer is either a number computed and compared against a threshold or a single entity identified by the evidence (a biomarker, or a binary safety or engagement call). The four lowest categories, target dependency (30%), differential response (28%), hit prioritization (19%), and cheminformatics (17%), instead require selecting a calibrated subset from many candidates with no unique answer, and average 24%. These tasks ask which genes are selective dependencies, whether a response is shared or lineage-selective, which screen hits to advance, and which substructures are genuine safety liabilities. The two groups differ by 34 percentage points (2.5-fold), and the ordering matches the failure taxonomy, in which tasks with a definite target are tractable while those requiring a calibrated cut under uncertainty are not.

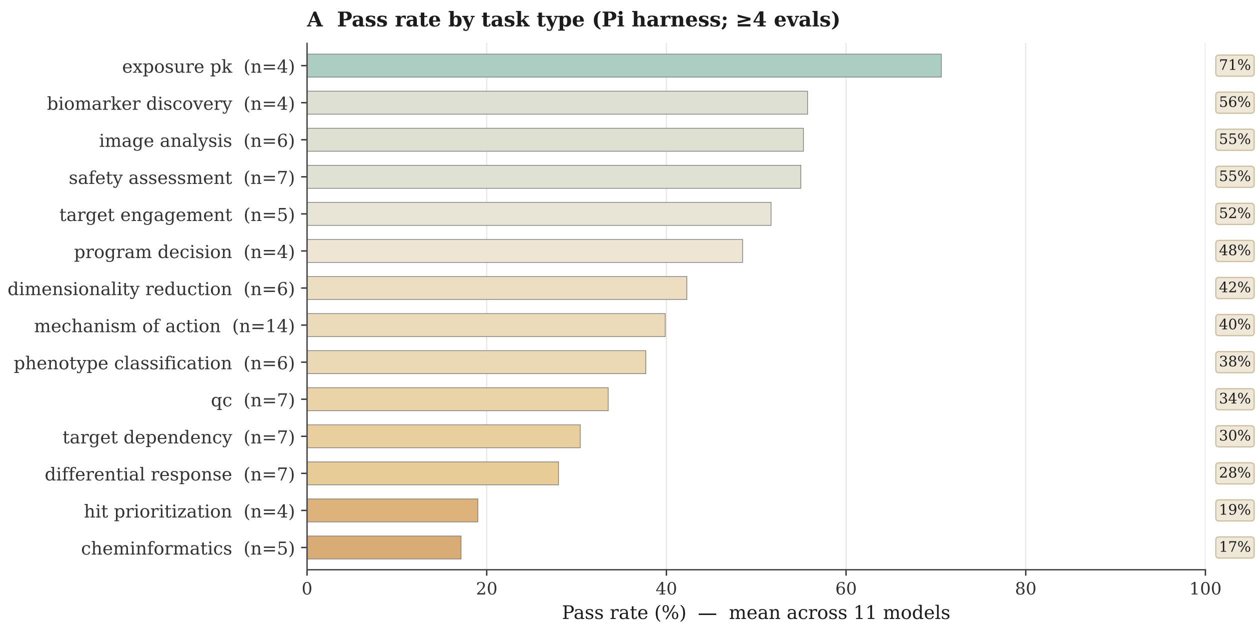



**Figure 6: Some task categories remain far harder than others.** Pi-harness pass rate by task type (categories with at least four evaluations), averaged across the 11 models. Difficulty tracks the kind of judgment required, from determinate quantitative tasks (exposure-PK) down to open-ended selection tasks (hit prioritization, cheminformatics).

Splitting performance by model and task type (Figure 7) shows that the two effects are largely separable. Within a model (a row of the matrix), overall model strength dominates, with the strongest models (claude-opus-4-8, gpt-5.5) scoring high across most categories and the weakest (grok-4.3, grok-4.20-reasoning) scoring low across all of them. Within a task type (a column), difficulty imposes a ceiling that a stronger model does not overcome. On the two hardest categories, hit prioritization and cheminformatics, no model exceeds 45%, indicating a limit shared across current models.

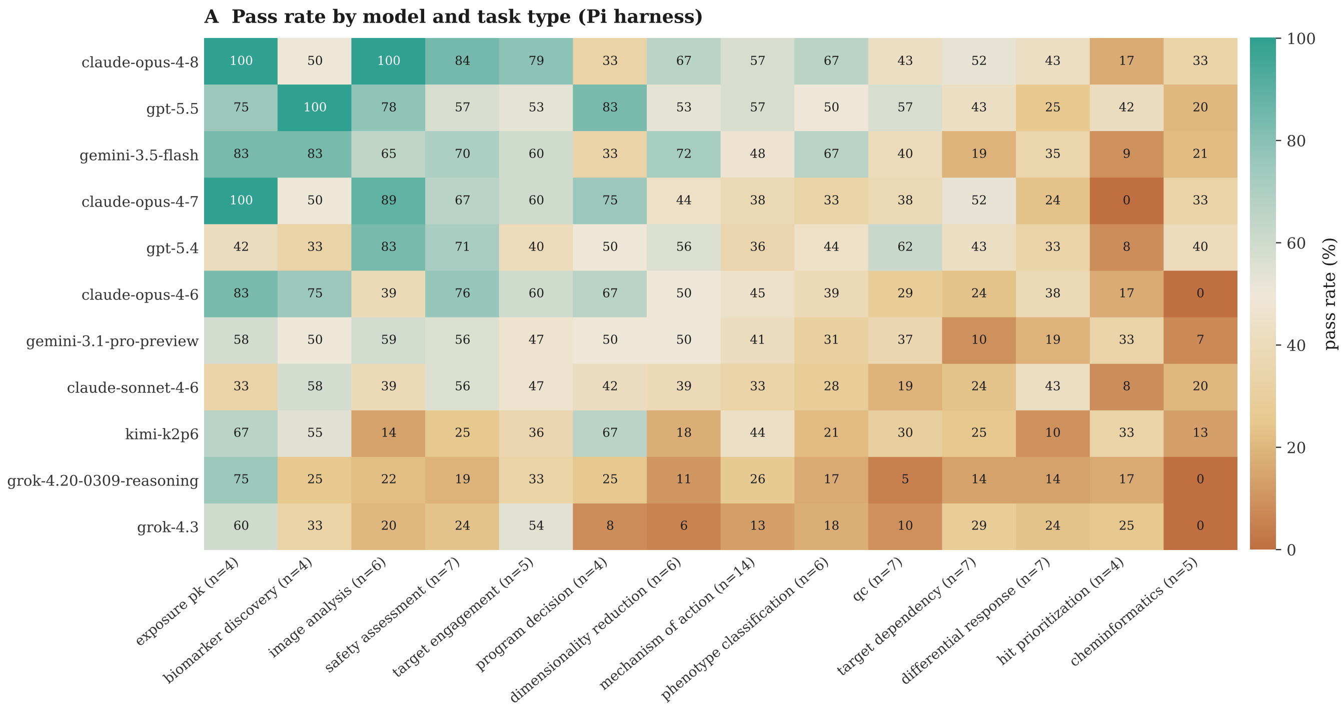



**Figure 7: Pass rate by model and task type.** Strict Pi-harness pass rate for each of the 11 models (rows, ordered by overall pass rate) across the task categories with at least four evaluations (columns, ordered easiest to hardest). Model quality sets the level across a row, but the hardest categories on the right stay low for every model.

## Among strong models, accuracy tracks process discipline rather than effort

Pass rate is only weakly and non-monotonically related to expended effort (Figure 8). Across the 11 models, mean agent steps and strict pass rate correlate at Spearman $\rho = 0.54$, dropping to $\rho = 0.32$ when the two low-engagement Grok models are excluded. grok-4.3 (mean \$0.02, median 6 steps) and grok-4.20-reasoning (\$0.11, median 11 steps) score lowest, at 18.3% and 19.7%, and under-investment drives 32% and 17% of their failures, against 0–1% for every other model (Figure 4C). Above the floor, effort and accuracy decouple. gemini-3.5-flash takes the most steps (median 43, ≈2.5× opus-4-8's 17) but reaches only 51.3%, below opus-4-8's 59.3%. gpt-5.5 beats gpt-5.4 (55.3% vs. 49.7%) with fewer steps (median 15 vs. 24). claude-opus-4-6 reaches 44.7% at a median of 10 steps, above claude-sonnet-4-6 (36.0%) and kimi-k2p6 (29.7%) despite their higher step counts.

Strong models differ mainly in the composition of their failures, not in their analytic ceiling (Figure 4C). The top Claude models fail almost entirely through engaged-but-incorrect analysis. Analytical errors account for 73–77% of their failures, leaving 23–27% non-analytical (claude-opus-4-6 cleanest at 23.0% of $n = 165$ failures, opus-4-7 at 26.8%, opus-4-8 at 26.7%). The GPT models reach comparable accuracy by a more process-laden route. Weak commitment (gpt-5.5 12%, gpt-5.4 16%) and abandoned-correct answers (17% and 15%) raise their non-analytical share to 39.7% (gpt-5.5, $n = 126$) and 41.8% (gpt-5.4, $n = 146$). Their headroom therefore lies in finalization, verifying the read and committing to the answer the analysis already supports, not in raw analytic capability. Under-investment is effectively absent from every strong model (≤1% of failures) and appears only in the two cheapest.

These quantitative patterns correspond to identifiable behaviors on individual tasks. The most accurate runs reason through the scientific intent, draw a calibrated cut, verify against the data, and commit to one answer. The pattern is clearest on hard tasks, where most runs fail but the strongest still succeed, and each hinges on a concrete drug-program decision. For instance, Bosutinib is an approved BCR-ABL/SRC inhibitor, but a program must credit its activity to the targets it actually engages in the chemoproteomics panel. On this task (32% of runs passing), claude-opus-4-8 overrode the clinical prior and committed to the non-canonical profile the panel supports, while other models anchored to the label returned the canonical ABL/SRC answer. The crizotinib control (21%) asks whether a separate broad-spectrum control compound justifies dropping genuinely engaged targets in a live-cell NanoBRET screen. It does not, and claude-opus-4-8 kept five engaged targets that other models discarded behind an invented quality-control gate, an error that wouldhave removed real hits from medicinal-chemistry follow-up. The EGFR/phospho-MEK task (44%) asks whether EGFR productively drives downstream MEK signaling in a BRAF-mutant colorectal-cancer organoid, the call that decides whether the compound advances as on-pathway. gpt-5.5 quantified the co-positive cell share in code rather than calling it from total marker abundance, reaching the supported conclusion.

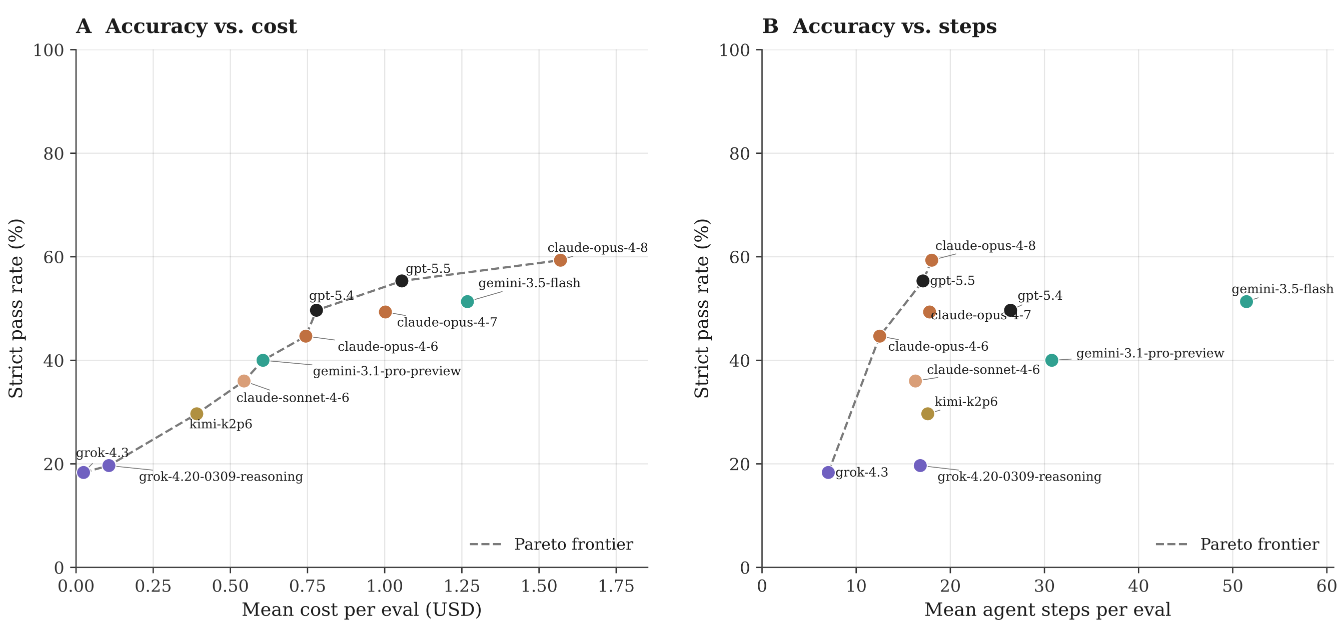



**Figure 8: Spending more does not buy accuracy.** Strict Pi-harness pass rate against mean cost (A) and mean agent steps (B) per evaluation (claude-opus-4-7 omitted, as its cost was not logged). The dashed line is the Pareto frontier. The cheapest, fewest-step models score lowest, but above a modest budget effort and accuracy decouple. The highest-step model (gemini-3.5-flash) is not the most accurate, and claude-opus-4-8 leads at moderate effort.

## Failures translate directly into wrong go/no-go decisions

The evaluations that most resemble a real pipeline gate are the program decisions, which hand the agent several assay readouts at once and ask for a single advance, hold, or prioritize call. The current suite has seven such advancement decisions, and across 230 runs models pass only 35% of them (Figure 9). Models perform well when the evidence package is clean and consistent, but struggle on more challenging gates, either advancing weak candidates or discarding strong ones.

*Clean correct.* The organoid program go/no-go is the clean case (Figure 9). Given a primary screen, a mechanism image, a tox counterscreen, and DMPK exposure for eight blinded compounds, the supported action is to advance the one compound that shows the desired cytostatic mechanism together with an adequate safety margin and covering exposure, and most models do (82%).

*Wrong synthesis.* On the decision-gate task the agent must assemble a set of models with convergent support across several endpoints, but models rank on a single synergy metric and nominate a model that is an artifact of that one number (27%), so the program's next experiments are built on the wrong models.

*False-go.* On the safety advancement task the supported call is that no candidate moves forward, because none pairs activity with healthy-tissue sparing, yet models advance one on activity alone or on a relative tumour-versus-healthy index (36%), putting an unsafe compound into the pipeline.

*False no-go.* The reverse error also appears. On the long-term combination task, models read a strong colony-stain reduction as a complete kill and drop a model the combination still supports (`HAI2027_FIG1_DUR01`, 48%).

Leading the benchmark does not predict making the better program call. Six multi-evidence advancement decisions (18 runs each) yields an order almost unrelated to overall accuracy (Spearman $\rho = 0.08$; Figure 10A). claude-opus-4-8, first overall at 60%, ranks last on these decisions at 22% (4/18). It clears both single-compound calls, advancing the one qualifying organoid compound (2/3) and holding when no candidate is safe (2/3), but it fails all four decisions that instead ask which preclinical models (cell lines and PDX models) should anchor the next program stage (the single-agent, priority, long-term, and decision-gate sets in Figure 10B), passing none of them (0/3 each) because it over-ranks on a single metric or carries too many candidates forward. gpt-5.5 is the only model strong on both kinds of task (15/18, 83%), committing to a defensible prioritized set rather than a full ranking. kimi-k2p6, ninth overall at 31%, rises to second here (12/17, 71%): its careful per-model computation pays off when the task rewards naming the supported models, though it still fails the restraint call by advancing a candidate on the no-go safety task (0/3). The two Grok models stay at the bottom (5/18 each) by answering with little of the multi-evidence work the decision requires.

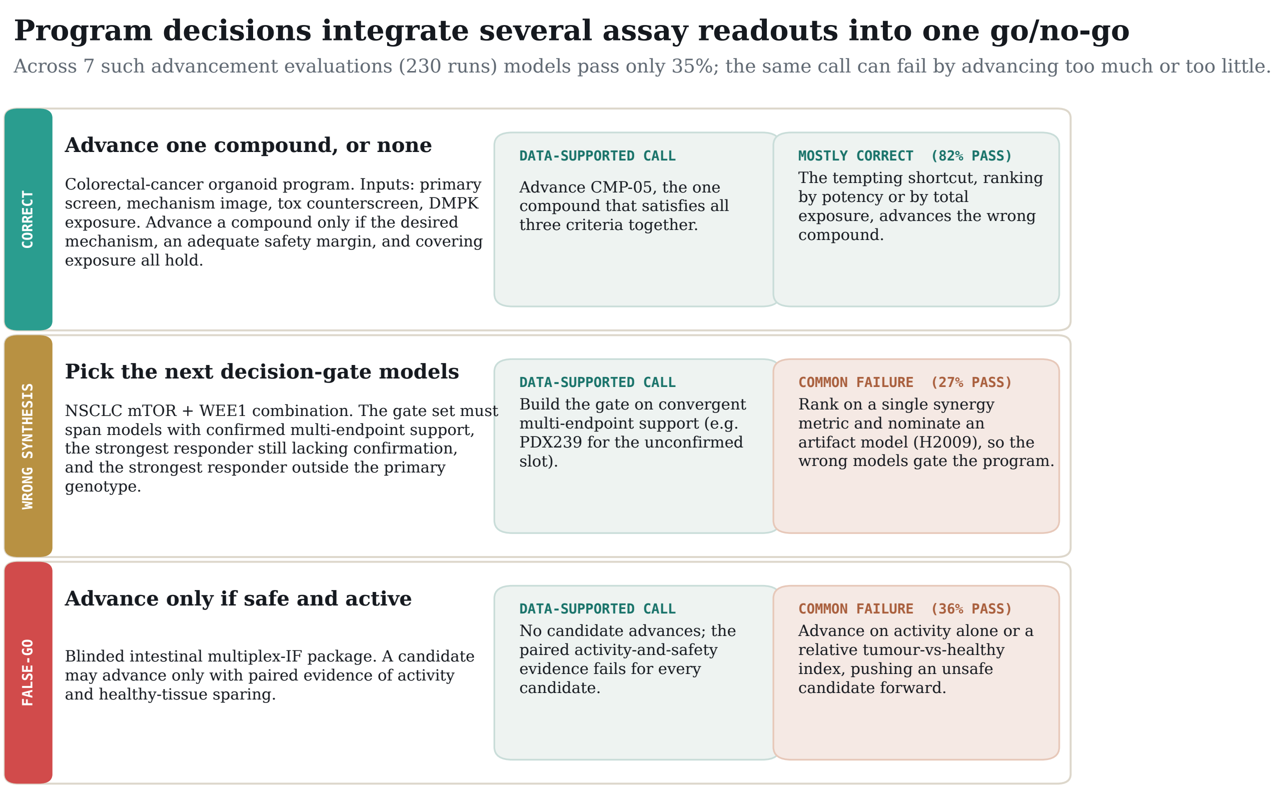


**Figure 9: Program decisions turn analysis errors into wrong go/no-go calls.** Three multi-evidence advancement evaluations, each giving the agent several assay readouts and asking for one advance, hold, or prioritize call. For each we show the evidence and the question, the data-supported call, and the common model error with its program consequence. The organoid go/no-go is passed by most models (82%); the decision-gate and safety-advancement tasks fail in opposite directions, advancing the wrong models (27%) or an unsafe candidate (36%).

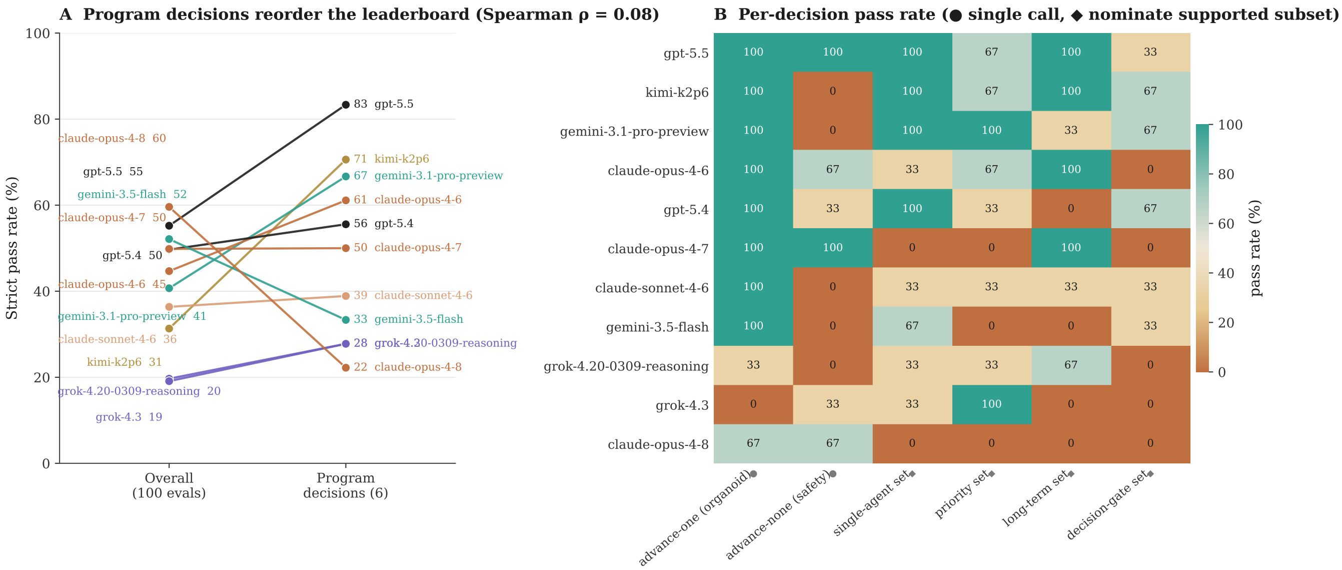



**Figure 10: Aggregate accuracy does not predict program-decision competence.** Pass rates on the six multi-evidence advancement decisions (Pi harness, 18 runs per model). **(A)** Strict pass rate on all 100 evaluations versus on the six program decisions; the integrative go/no-go calls reorder the leaderboard almost completely (Spearman $\rho = 0.08$), with the overall-best model, claude-opus-4-8, falling to last and kimi-k2p6 rising from ninth to second. **(B)** Per-decision pass rate by model. The two single-compound calls (advance one, or hold; marked ●) are passed broadly, but the four tasks that require nominating the supported subset of models (marked with a diamond) separate the field; claude-opus-4-8 fails all four.

## The agent harness has a surprising impact on model performance

Although a harness serves merely as a wrapper around a fixed model, utilizing the same checkpoint, task text, and answer-submission protocol. We found that it exerts a surprisingly large and consistent effect on overall performance. In within model matched comparisons, which provided the only unconfounded contrast since each harness evaluated a distinct set of models, Pi outperformed Claude Code by 4.4 percentage points (95% CI: 1.2–7.8) on the shared Opus models. Similarly, Pi outperformed OpenAI Codex by 5.5 percentage points (95% CI: 1.2–9.8) on the shared GPT models. Notably, Pi achieved a higher score on every shared model, rather than merely winning on average.

The magnitude of this improvement is comparable to a model-generation upgrade. For instance, transitioning from Opus version 4.6 to 4.7 yielded only a 4.6-point increase on Pi. Consequently, the choice of harness is roughly as consequential as upgrading to a newer model version. Furthermore, this advantage was relatively uniform across both image-based and non-image tasks, indicating that the performance gains were not primarily attributable to visual access. For context, Claude Code rendered figures to the model in 100% of image-bearing tasks compared to 84% for Pi, whereas OpenAI Codex, which lacks vision capabilities, never rendered figures.

Instead, the residual performance differences can be traced to subtler scaffolding properties. For example, Pi's lean, three-tool loop failed on only 0.6% of runs—compared to 1.8% for Claude Code, and reached an answer in fewer steps. Conversely, OpenAI Codex's lack of a vision pathway further penalized its performance on figure-decisive tasks. Although these effects are modest in absolute terms and present wide confidence intervals, they are directionally robust and persist across different model families. Ultimately, an implementation detail nominally orthogonal to the model itself can shift the endpoint pass rate by a margin equivalent to a full model generation.

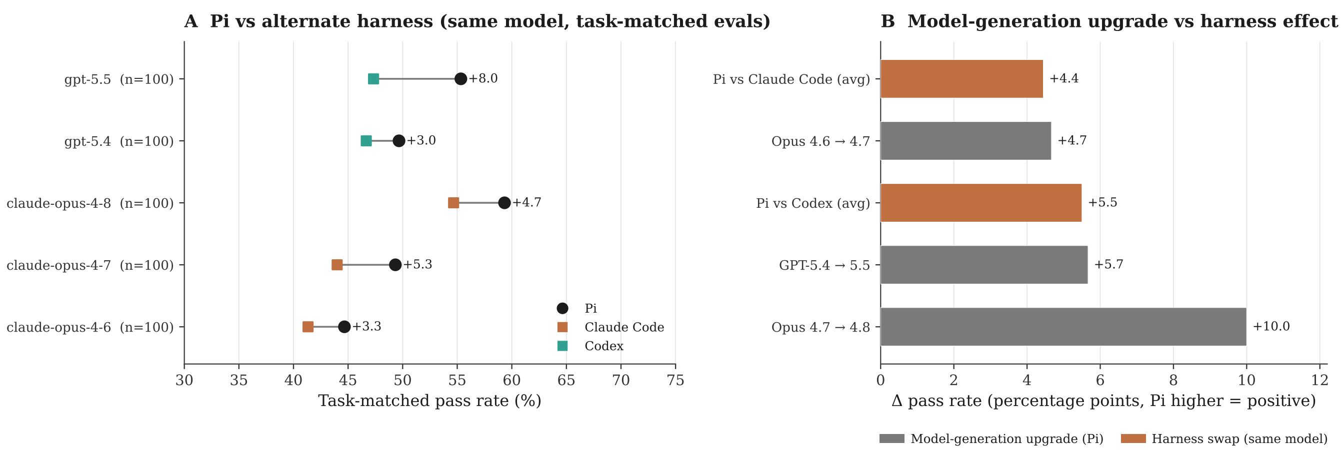



**Figure 11: Harness choice is as consequential as a model-generation upgrade.** (**A**) Task-matched pass rate on Pi versus the alternate harness for each model pair. Points are matched on the shared evaluation set; the labeled gap is the within-model Pi advantage in percentage points. (**B**) Head-to-head comparison of the Pi advantage over an alternate harness (terracotta) against the gain from a model-generation upgrade on Pi (grey). A harness substitution shifts pass rate by a margin comparable to upgrading one model generation, so reported scores should be interpreted as a property of the model-plus-harness, not the model alone.

# Discussion

TxBench-PP measures a narrow but important part of drug discovery: whether an agent can turn preclinical pharmacology evidence into a defensible local decision. This scope is intentionally smaller than end-to-end therapeutic discovery. It is a small-molecule cluster benchmark within the broader TherapeuticsBench roadmap, not a claim to cover every stage or modality. It does not yet test target genetics, full disease-model selection, primary screening operations, broad clinical translation, or non-small-molecule modalities. The narrower scope makes the benchmark verifiable while still covering decisions that directly affect compound advancement.

The central hypothesis is that current agents will often operate the tools of preclinical analysis but fail at the scientific judgment layer. If a model selects a mechanism from a drug label, trusts a contaminated target-engagement signal, ignores survivor bias, or ranks by total exposure rather than unbound coverage, it may produce a fluent and computationally elaborate answer that is not the pharmacology decision supported by the data.

[**TODO:** After results are available, add a paragraph comparing endpoint pass rates, partial-credit or field-level scores if available, and trajectory review. Emphasize what capability is missing and which tasks show meaningful progress.]

Future versions can extend the benchmark upstream to target rationale and primary screening, sideways to pharmacogenomics and disease-model characterization, and across modalities to antibodies, ADCs, degraders, oligonucleotides, cell therapies, and gene therapies. A complementary long-horizon TherapeuticsBench track can follow program stories across discovery, development, and translation. For the current paper, the clean claim should remain: TxBench-PP is a verifiable cluster benchmark for small-molecule preclinical pharmacology decisions.

# Methods

## Task construction

Candidate evaluations are derived from realistic preclinical pharmacology workflow states. A task is retained when the target conclusion can be recovered from the supplied files, expressed through a constrained final answer, and graded deterministically. Tasks should be revised or removed when the prompt over-specifies the method, when multiple reasonable analyses produce incompatible decisions, or when the grader cannot distinguish the supported pharmacology conclusion from a plausible but unsupported answer.

## Agent runs and aggregation

[**TODO:** Insert model-harness list, number of attempts per evaluation, execution environment, parsing rules, and how invalid or missing outputs are counted.]

## Statistical analysis

[**TODO:** Insert primary metric and confidence interval convention. If following SpatialBench/EpiBench, report evaluation-weighted endpoint pass rate, raw pass counts for interpretability, and Student $t$ confidence intervals over evaluation-level replicate means.]

# Data availability

[**TODO:** Insert public repository, public example evaluations, result summaries, and trajectory availability language.]

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